Open Questions

Playmaker rules engine · commit 2f66076 · generated 05 Nov 2026

Every ruling the rules leave open is a named setting in `playmaker_engine/rules_config.py` (class `Rules`). Changing a ruling is one line: `with_rules(setting=value)` or edit the default. Items 1 to 12 come from the build prompt. Items 13 onward were found while building or reading the PDFs. Section C lists where the reference PDFs and the prompt disagree.

Designer rulings (3 Nov 2026). Ruled and built: 5, 12, 16, 17, 20, 27, 32, C1, C4, plus a new rule, the and-one (34). Confirmed as built: 1 to 4, 6, 8 to 11, 13 to 15, 18, 19, 21 to 26 and 28 to 31. Item 33 follows from the and-one. Still open: 7 (see 38), re-tested by the solver at full budgets. Ruled in step 2b: 35. Closed in step 2b: 39. Rows marked **RULED** show the ruling.

A. Questions from the prompt

#	Question	Default (setting)	Alternative	What it changes
1	Ball after an intercept or steal that "plays on"	Ball back to the passer (intercept) or stays with the carrier (steal), −5, play continues. `turnover_play_on='play_on'`	`'end'`: the possession ends	Whether a turnover is a setback or a full stop. The Offence tracker ("Fail: no pass, −5") and the rulebook score table ("−5, play on") both support the default.
2	When a pass resolves	On the defence's next turn. It may Intercept; if it does anything else, the pass lands at the start of that turn. `pass_resolution='defence_turn'`	`'immediate'`: lands at once unless an on-line defender is already there	Whether arrow-across defenders get a chance to close the lane. Under the default an inbound pass that lands mid-turn starts the new phase at once, so the defence's turn ends there.
3	Sprint length	Up to 3 + QS straight, 2 + QS diagonal. `sprint_orth_base=3`, `sprint_diag_base=2`	Any other base	Reach of fast players.
4	Roll bounds	Target ≥ 20 always succeeds, ≤ 0 always fails, no naturals. `natural_rolls=False`	`True`: natural 1 always succeeds, natural 20 always fails	Sure things (Pass 20 v Intercept 0) and hopeless rolls.
5	Negative value on a declared shot	**RULED: floor at 0**, matching the forced shot. `floor_declared_shot=True`	`False`: scores as negative	Otherwise declaring a bad shot could be worse than letting the clock run out.
6	Defence reaction to a declared shot	0, 1 or 2 actions from tokens still in the pool: Move (incl. arrow turn), Quick Step, Sprint, Steal. Steal limits apply. Empty pool: only the contest. `reaction_kinds`, `reaction_may_decline=True`	Force at least one action; widen the kinds	How much a declared shot can be closed out. Declining is allowed because the contest is automatic and forcing a pointless move adds nothing.
7	Defence out of tokens while the offence has some	Phase ends at once, as written. `defence_empty_ends_phase=True`	`False`: the offence plays out its tokens	Whether the defence can "run out the phase" by spending fast. **Open: the solver will show whether the defence exploits it.**
8	Hesitate / Pause chaining	Each passes the turn and uses a token. Pause is illegal for the whole defence turn right after an offence turn that ended with Hesitate, and vice versa. `wait_chain_block=True`	`False`: no chaining limit	Stops wait-for-wait stalling. **Needs confirmation** of "straight after" = the whole following turn.
9	Screen geometry	A defender's diagonal step is banned if it starts on the T square, or cuts the T square's corner (T is one of the two squares the diagonal passes between). `screen_rule='from_t_or_corner'`	`'from_t_only'`	Exact squares are in `tests/test_movement.py::test_screen_t_square_exact_squares`: screener I6, T on I5. From I5 a Quick Step reaches only J5, I4, H5 (H4, J4, H6, J6 banned). From H5, I4 is banned (cuts T's corner); H6, I5, H4, G4, G5, G6 are open.
10	Arrow line crossing	Illegal only if the carrier's path goes from one side of the ray to the other (touching it, or stepping onto it and back, is fine). Checked over the whole path of one movement action. `arrow_line_strict=True`	`False`: stepping onto the ray at all counts	How hard an adjacent defender walls off a side. A carrier who ends one action on the line and leaves to the far side with the next is not caught (each action is checked on its own).

#	Question	Default (setting)	Alternative	What it changes
11	Move path order for 1 straight + 1 diagonal	Either order, if the middle square is empty. `move_either_order=True`	`False`: straight first (the old script)	Explains several regression differences (REGRESSION.md).
12	Contest choice	**RULED (step 3b): the player decides.** After its reaction the defence chooses: contest with any adjacent defender (clean with the arrow on the shooter, or a foul roll with it off), or don't contest. `contest_choice='player'`; the solver's best choice equals the old automatic minimum (tested). Earlier ruling: the defence chooses any adjacent defender (fix the rulebook's "nearest" wording, C1). The defence picks the adjacent defender, or nobody, with the lowest expected value, counting foul free throws. `contest_choice='min_value'`	`'nearest'`: the nearest adjacent defender (rulebook wording, see C1)	Note: under the default the defence sometimes prefers an arrow-off contest because free throws (2 × FT) are worth less than a great look. That is a real "foul the shooter" decision the rules allow.

B. New questions found while building

#	Question	Default (setting)	Alternative	What it changes / source
13	When is a defender "on" the pass line?	Its square's centre lies within 0.5 of the segment, strictly between passer and receiver. `on_line_tolerance=0.5`	`0`: exact centre on the line (as `check_scenarios.py` uses for deny arrows); or any overlap with the square	How many squares can intercept directly. Prompt "Intercept"; Defence tracker "On the line: any arrow".
14	Quick Step length	Up to the card value (a QS-2 player may step 1). `qs_up_to=True`	Exactly the card value	Defence tracker "A burst set by the card: 2, 1 or 0 squares".
15	Sprint with Quick Step 0	Not allowed. `sprint_needs_qs=True`	Allowed (it is then a straight Move that spends a Quick Step)	Only removes a dominated option.
16	Who may Challenge a screen	**RULED: only the defender standing on the screen's T square** (the one it blocks; covers the post seal). `challenge_who='on_t'`	`'adjacent'`: any defender next to the screener; `'any'`	Defence tracker "Name the screen you're fighting through".
17	Can a locked ball carrier Dribble?	**RULED: no.** A lost dribble means pass or shoot. `locked_can_dribble=False`	Yes	Offence tracker "Lose: you're locked, but can still pass or shoot" could be read as listing the only allowed ball actions.
18	How long a Dribble win frees the carrier from that arrow line	Rest of the phase (the defender's lock clears then too). Tied to that carrier: after a pass the new carrier is bound again. `dribble_win_lasts='phase'`	Until the carrier's next movement action	Offence tracker "Win: they're locked; you may cross their arrow line".
19	What the inbounder may do before the ball is in	Pass only (no Dribble, Screen or movement). `inbounder_pass_only=True`	Normal offence actions except movement	Scenario rules "It cannot move". Defenders may still Steal from an inbounder whose arrow is on it (normal Steal rule).
20	Can the far-contest −3 take a contest below 0?	**RULED: no, floor at 0.** A contest never makes a shot better. `floor_contest_zero=True`	Allow below 0	Prompt "the contest is −3".
21	A side with tokens but no legal action	It forfeits its remaining tokens (defence: phase ends; offence: defence gets its final turn). `forfeit_when_stuck=True`	Pass the turn without a token	Rules are silent; needed so every game terminates.
22	Need a Three: free throws from a fouled two, and steal-foul free throws	Score 0 (only threes count). `three_only_ft_on_twos=False`	They count	Scenario rules "only a three-point shot scores".
23	Coach's Call: which band, and does a forced shot count?	Band of the look (shot value, or the free throws if fouled). A forced shot by the target counts. `coach_bonus_on_forced=True`	Band of the possession total; forced shots excluded	Scenario rules "add the bonus for how good the look was".
24	Which defenders bind the carrier's arrow line	Those adjacent at the start of the movement action. `arrow_line_adjacent_at='start'`	Adjacent at any point of the path	Prompt "adjacent defender".
25	"Up to 2 diagonal"	A straight diagonal line. `diag_move_straight=True`	Any two diagonal steps (e.g. NE then NW)	Trackers "or 2 diagonal".
26	Steal-foul free throws when no shot is declared	2 × the ball carrier's FT rating. (engine behaviour)	—	Rulebook "Free throws score 2 × the shooter's FT rating" names a shooter; for a steal foul there is none.
27	Inbound not in after two phases: tie for "furthest from the rim"	**RULED: the offence chooses.** A new offence decision (`pick`) among the tied players. `inbound_tie='offence'`	`'lowest'`: lowest piece number	Scenario rules "offence picks if tied". The off-court inbounder is excluded.

#	Question	Default (setting)	Alternative	What it changes / source
28	Where the inbounder goes once the ball is in	It stays on its off-court square until it moves; its movement steps onto the court. It can receive a pass there but can't shoot (no zone). (engine behaviour)	Place it on the edge square automatically	Scenario rules "the inbounder may step onto the court".
29	Re-screening and screener movement	A screener can't Screen again until it flips back, and can't Quick Step or Sprint either ("can't move"). Screens survive the phase reset and flip back after the offence's next turn. (engine behaviour)	Allow a refreshed screen	Offence tracker "Can't move; flips back".
30	Forced-shot type	**RULED (step 3b): the player decides.** The offence picks the type its zone allows at clock 0, then the defence chooses the contest. `forced_shot_choice='player'`	`'best'`: the engine picks	Rulebook "the ball carrier takes a forced shot".
31	Arrow-across defender who moved onto the line but then doesn't Intercept	Allowed: the pass then lands. (engine behaviour)	Must Intercept	Prompt "must first reach the segment".
32	Lock token supply	**RULED: no cap.** At most 2 Dribbles and 2 Challenges per phase means at most 4 new locks; Star Locked can reach 5 in phase 1, and the printed token sheet has a spare.	Cap at 4	Rulebook p1 "4 lock tokens".
33	Two or Three with free throws	A fouled three with 30+ in foul points counts as an Excellent three (win). (engine behaviour)	Only made shot values count	Scenario rules "an Excellent three wins".
34	**New rule: the and-one**	**RULED.** Once a shot is declared, or forced at clock 0, any foul (a contest foul, or a Steal foul during the reaction) scores the **higher of the free throws and the shot's uncontested value** (points × rating). `and_one=True`	`False`: free throws only	Fouling a good look can never help the defence. Checked pair by pair: 0 of 39,375 offence × defence × shot-type pairings break it; without it, 10,725 do. A side-on closeout pays in 72.1% of pairings; see REGRESSION.md. Steal fouls before a shot is declared are unchanged.

C. Where the reference PDFs and the prompt disagree

None of these block the engine; each is a setting or a wording question for the designer.

1. **Who contests.** *Ruled: the defence chooses; change the rulebook wording.* `Playmaker_Rulebook.pdf` p1, "The shot": "Contest: the nearest defender's rating against it." p2, "Shooting": "Then the nearest defender contests." The prompt and both trackers (`Defence_tracker.pdf`, CONTEST: "Then one adjacent defender contests") let the defence choose among adjacent defenders. Default follows the prompt; `contest_choice='nearest'` follows the rulebook.
2. **Sprint and Quick Step.** `Defence_tracker.pdf` / `Offence_tracker.pdf`, SPRINT: "Uses that player's Move and Quick Step." The prompt says it "counts toward Quick Step" (the side's limit of 2). The tracker wording suggests a per-player Quick Step; the limit column says "×2 Sprints count" (per side). Engine: per side, as the prompt says.
3. **The adjacent defender on a pass.** `Defence_tracker.pdf`, INTERCEPT: "On the passer: only a pass through you." `Offence_tracker.pdf`, PASS: "Your defender: only their arrow counts." The two describe the same defender differently (arrow pointing at the passer vs. any defender of the passer). Engine follows the prompt's merge: an adjacent defender whose arrow points at the passer can only stop a pass through its own square; any other adjacent defender counts only if its arrow crosses the pass (and must move onto the line first).
4. **What a lock means.** *Ruled: change the rulebook to "can't Move or Sprint".* `Playmaker_Rulebook.pdf` p1, components: lock tokens "Mark a player who can't move this phase." The trackers (LOCKED) and the prompt allow Quick Step while locked. Engine follows the trackers.
5. **Plain wood name.** `Offence_tracker.pdf`, SHOT: "never from plain maple"; everywhere else "plain wood". Wording only.
6. **Scenario 10 card text** (`Playmaker_Scenarios.pdf`, Late Clock Reset): "Both Screens are spent: O4's is down, O5's is still up. Offence has 4 tokens left, defence 4." The prompt only says "tokens already spent and O5's screen standing". Engine uses the card's numbers (4 and 4, Screen limit used).

Nothing in the PDFs contradicts the prompt's numbers (6 tokens, 2-second phases, −3, −5, +5, bands, bonus ladder, roll rule, free throw multipliers).

D. Found by the solver (step 2)

Evidence is in `SOLVER_REPORT.md` and `solver_results/runs.json`. Priority labels follow the designer's order (System Fidelity > Neutrality > Player Agency > Legibility > Accessibility). No rule was changed.

#	Finding	Evidence	Affects	Options for the designer
35	**RULED (step 2b): an inbound pass works like any other pass.** `inbound_resolves='after_defence_turn'`: the defence gets its normal turn (1 or 2 actions: Move then Intercept, Sprint then Intercept, or move two players) and the pass resolves after it; then the phase resets to the card's clock, offence first. Found: a completed inbound pass gave the receiver a free turn. Under OQ-2 the pass lands at the start of the defence's turn, and the scenario rules then reset to a new phase with the offence first. The defence's turn is skipped, so the receiver gets a Move and a Shot against only the reaction.	Baseline Box: the defence's best line is to let the pass land (End turn), then O1 Sprints and shoots: 36, a Gift. Two setup fixes only cut it to 30 (still a Gift).	Fidelity, Neutrality	(a) After the ball is in, the defence acts first in the new phase. (b) The inbound pass resolves at the end of the defence's turn instead of the start (OQ-2 alternative for inbounds only). (c) Keep it and accept that inbound plays are about the catch.
36	**Pause appears in 10 of 16 principal lines** (the defence's most-used action).	Rule health 2. Typically the defence answers a pass or a drive by doing nothing.	Fidelity (possibly a solver effect)	Partly a solver effect: the evaluation does not score defensive positioning beyond the next shot, so standing still costs nothing. Widen the evaluation before reading it as a rule problem. If it persists, Pause may be too cheap a way to make the offence commit.
37	**Challenge, defensive Sprint closeouts and deliberate fouls never appear.** Steal appears on 2 lines, Hesitate on 1.	Rule health 2 and 3.	Fidelity	Challenge: only the defender on the T square may Challenge (OQ-16), and the solver's lines rarely contain a standing screen. Sprint closeouts lose to Move-then-Quick-Step (arrow on, no foul roll), which matches the and-one design. No foul outcomes on any line.
38	**Re-tested at full budgets (step 2b): no effect. Designer ruling needed to close.** OQ-7 (defence's last token ends the phase).	16 cards at 2 and 10 minutes, phase ends as written v phase runs on: 32 comparisons, **no value moved**; 3 openings changed at equal value (cards 8, 9, 10 at 10 minutes).	Neutrality	The defence never spends fast to end the phase under best play. Recommendation: keep as written (`defence_empty_ends_phase=True`) and close.
39	**CLOSED (step 2b): keep as built.** Clock expiry never happens under best play.	Every principal line shoots inside the horizon. A forced shot from the cards' starting positions is worth 0 on 12 of 16 cards (the carrier starts on plain wood), 14 to 19 on the others.	Neutrality	Because forced shots are floored at 0, "forced shot or −5, whichever is worse" is always just −5: the alternative removes the forced shot.
40	**Verdicts at full budgets (step 2b).**	At 2 and 10 minutes, 5 seeds: 4 of 16 cards unstable (spread over 2 points or the opening changes between budgets), 12 estimated, none exact. Most cards reach only 2 turns deep at 10 minutes.	Legibility of the results	Verdicts are firmer than in step 2, but read them as "best over the next few turns".

E. Solver round 2 (step 2b)

#	Finding	Evidence	Affects	What to decide
41	**The one-turn evaluation was blind to defensive position.** It scored only the offence's next shot, so a defender standing still cost nothing (the Pause result). The evaluation now looks over the offence's next two turns against the defence's best reply.	Agreement with the exact endgame solver on 50 positions: mean error 5.38 → 0.61, within 2 points 46% → 92%. Positional terms (open lane to a shooter, carrier on a shooting zone) did not improve agreement (0.65, 1.30) and are off.	Fidelity of every solver result	None: done.
47	**Pause is the most-used defensive action in the principal lines, but it is a search artefact.**	Principal lines at 10 minutes: Pause 10 of 22 defence steps. Searched on its own, each example position has a Move clearly better than Pause: Need a Three 26.0 v Pause 30.0; Sideline, Late 15.0 v 27.2; Late Clock Reset 15.0 v 19.5. Pause appears where the defence's turn sits below the root of the search: there only concept plays are tried (raw actions are added at the root only) and little depth is left.	Fidelity of every verdict	No rule change. The same shallowness may flatter the offence, so Gift verdicts are provisional; the next solver task is wider defence candidates below the root.

#	Finding	Evidence	Affects	What to decide
48	**Dead rules: Challenge Screen and defensive Sprint closeouts.**	Across 179 best lines (every card, budget, seed, target and matchup): Challenge 1 step, defensive Sprint 0. The offence's best lines use a Screen in only 5 of 179, so there is rarely a screen to challenge; screen-heavy teams on the screen cards did not change it.	Fidelity	**Designer ruling:** are screens too weak to set (so Challenge never matters), or is Challenge too costly? The search reaches 2 turns on most cards, so a longer horizon could still find them.
49	**The Coach's Call secret is worth nothing at full budget.**	Cards 8, 9, 11 at 10 minutes: the defence's best single strategy without knowing the target holds the offence to exactly its known-target value (20.5, 26.0, 28.8).	Player agency	**Designer ruling:** the hidden target adds no value against best defence. Raise the bonus, or let the target's identity change something the defence must commit to early.
50	**Step 3d team rebuild changed one verdict.**	Re-solved with the new Twin Towers and Ball Movement: Baseline Stack Wall (18.0) → Puzzle (16.6); the other five keep their verdicts (values in the Solver Report).	Results only	None: recorded so the old verdict is not reused.
51	**A locked star can still Quick Step, so Star Locked's one plan is the star himself.**	Star Locked at 10 minutes: O1 (locked) Quick Steps to H10 and shoots the three (24.0, 1 plan). By the ruling that a lock means "can't Move or Sprint", a Quick Step is allowed. With other matchups the plan changes to a kick (someone else scores).	Fidelity, the card's lesson	**Designer ruling:** should a lock also stop Quick Step? If the card's lesson is "someone else has to score", yes; otherwise the setup fix (X1 starts on H10) makes it a Puzzle.

F. Playtest build (steps 3 and 3b)

Legibility questions raised by the interface; none changes a rule.

#	Question	Status
42	Who rolls. Opposed rolls are the offence's; foul rolls are the defender's straight roll.	Every roll now says "Offence rolls" or "Defence rolls" (preview, last roll, log).
43	Opening placement.	Built for full games, with the preview setup as a shortcut and a documented computer rule.
44	Blocked diagonal steps around a screen just don't appear as options.	Open: the interface does not yet say why a square is missing.
45	Coach's Call in hot-seat relies on the defence looking away.	Open (remote play hides it properly).
46	Court squares on a phone are about 22 px, under the 44 px touch target.	Every court choice is repeated as a full-size button on phones.